

OPEN-VOCABULARY OBJECT DETECTION VIA VISION AND LANGUAGE KNOWLEDGE DISTILLATION

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Outline

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Background: Vision–Language Models Unlock Flexibility

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Background: Computer Vision is Transforming the World

- Vision powers autonomy, robotics, retail, AR, and content moderation.
- Object detection is a core capability for perceiving and interacting with the world.
- Real-world vocabularies are vast and long-tailed; static label sets quickly become outdated.
- Web-scale image-text learning is enabling open-vocabulary recognition.

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Background: Vision–Language Models Unlock Flexibility

- Abundant web image–text pairs enable large-scale contrastive training (e.g., CLIP, ALIGN).
- Aligned text/image embeddings allow zero-shot classification via prompts.
- However, object-level detection adds localization, fine-grained recognition, and efficiency constraints.
- Bridging image-level generalization to object-level is still challenging.

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Problem: Open-Vocabulary Object Detection

- Detect objects described by arbitrary text, including categories unseen during training.
- Instance-level annotations are costly; long-tail categories have sparse labels.
- Need detectors that generalize beyond fixed label sets using text prompts.

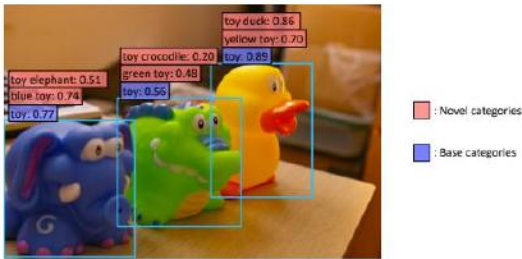


Figure: Open-vocab detector with text: base-trained, novel classes

Why This Is Hard

- Detection labels (boxes/masks) do not scale easily to thousands of classes.
- Typical detectors use class-specific heads tied to fixed vocabularies.
- Naïve CLIP-on-crops is slow and brittle to distortions and clutter.
- We need accuracy, speed, and plug-and-play vocabulary updates.

Core Idea: Distill Vision–Language Knowledge into a Detector

- Use a pretrained open-vocabulary classifier as teacher (e.g., CLIP/ALIGN).
- Align detector region embeddings with teacher text and image embeddings.
- Keep localization class-agnostic; swap text embeddings at inference for any classes.

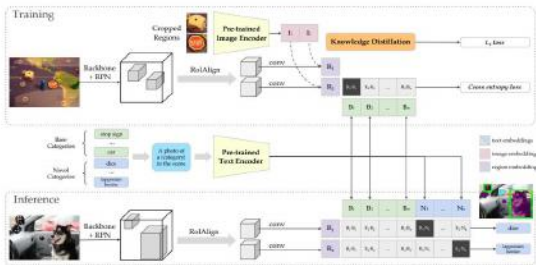


Figure: ViLD开放词汇检测：蒸馏预训开放词汇分类模型，以文本嵌入作区域分类器并最小化区域与图像嵌入距离，推理时支持新类别文本。

Contributions at a Glance

- ViLD: Distill text (classifier replacement) and image (visual feature) signals into a two-stage detector.
- Fast open-vocabulary inference; optional ensembling to balance base vs. novel categories.
- State-of-the-art zero-shot/generalized detection; direct cross-dataset transfer by swapping texts.
- Analyses: proposal generalization, failure modes, prompt and loss ablations.

- Two-stage detector (e.g., Mask R-CNN) with class-agnostic box/mask heads.
- ViLD-text: fixed text-embedding classifier + learnable background; cosine logits.
- ViLD-image: L1 distillation of region embeddings to teacher's crop embeddings.
- ViLD combines both objectives to learn an open-vocabulary head.

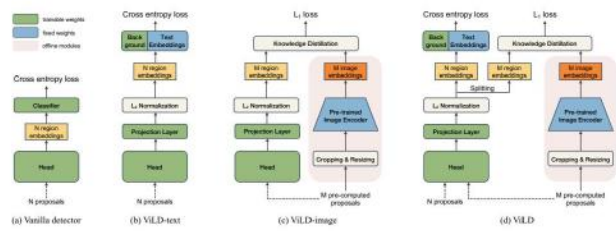


Figure: ViLD架构: (a)二阶段基线; (b)以文本嵌入+背景替代分类器并投影对齐; (c)L1蒸馏预计算图像嵌入; (d)联合。

- Generate category embeddings with teacher's text encoder; use prompt ensembling/synonyms.
- Replace learnable classifier weights with normalized text embeddings.
- Train region features to maximize cosine similarity to the correct text embedding.
- Include a learnable background vector; apply temperature-scaled cross-entropy.

- Precompute teacher image embeddings on proposal crops ($1.0\times$ and $1.5\times$ context).
- Align region embeddings to teacher crops using L1 loss (better than L2 in AP_r).
- Offline proposals and renormalization boost efficiency; expose novel visual concepts.
- Distillation complements text alignment to stabilize open-vocab learning.

- Total loss = detection losses + text CE + $\lambda \times$ L1 distillation.
- λ trades off base vs. novel performance; tuned per deployment needs.
- Inference: swap any set of text embeddings; classify with cosine similarity; per-class NMS.
- Enables interactive queries and systematic vocabulary expansion without finetuning.

ViLD-ensemble: Separate Heads, Better Balance

- Training objectives can compete (text vs. image); split into two heads.
- Train text and image heads separately; ensemble scores at inference.
- Weighted geometric mean leverages complementary strengths (base vs. novel).
- Improves overall AP without extra runtime vs. single-head ViLD.

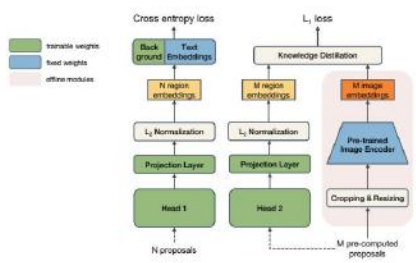


Figure: ViLD-ensemble架构与训练目标: 同ViLD, 采用两个同构头以降低文本/图像目标竞争, 推理时两头结果集成。

Experimental Setup: Datasets, Metrics, Protocols

- LVIS v1: 866 base, 337 rare (novel); primary metric: mask APr (rare).
- COCO open-vocab (48 base/17 novel), VOC 2007, Objects365 for transfer.
- Zero-shot transfer: train on LVIS base, test elsewhere by swapping text embeddings.
- Baselines: supervised (base/novel), RFS, CLIP/ALIGN on cropped proposals.

Result: Proposals Generalize to Unseen Categories

- RPN trained on base-only achieves near-identical AR on novel categories.
- ≤ 2 AR point gap vs. base+novel indicates strong class-agnostic proposal quality.
- Supports using class-agnostic localization for open-vocab detection.

Table: Table 1: Training with only base categories achieves comparable average recall (AR) for novel categories on LVIS. We compare RPN trained with base only vs. base+novel categories and report the bounding box AR.

Supervision	ARr@100	ARr@300	ARr@1000
base	39.3	48.3	55.6
base + novel	41.1	50.9	57.0

Core LVIS Performance: ViLD vs. Baselines

- ViLD (R50) achieves 16.1 APr, surpassing Supervised-RFS by +3.8 APr.
- ViLD-ensemble further improves overall AP without extra runtime.
- Stronger teachers scale performance: ViT-L/14 and ALIGN yield large APr gains.

Table: Table 3: Performance of ViLD and its variants.

Backbone	Method	APr	APc	APf	AP
ResNet-50+ViT-B/32	CLIP on cropped regions†	18.9	18.8	16.0	17.7
	ViLD-text+CLIP†	22.6	24.8	29.2	26.1
	Supervised-RFS (base+novel)	12.3	24.3	32.4	25.4
	GloVe baseline	3.0	20.1	30.4	21.2
ResNet-50	ViLD-text	10.1	23.9	32.5	24.9
	ViLD-image	11.2	11.3	11.1	11.2
	ViLD (w=0.5)	16.1	20.0	28.3	22.5
	ViLD-ensemble (w=0.5)	16.6	24.6	30.3	25.5
	ViLD-ensemble w/ ViT-L/14 (w=1.0)	21.7	29.1	33.6	29.6
	ViLD-ensemble w/ ALIGN (w=1.0)	26.3	27.2	32.9	29.3
EfficientNet-b7		26.3	27.2	32.9	29.3
ResNeSt269+HTC	2020 Challenge winner (Tan et al., 2020)‡	30.0	41.9	46.0	41.5

Qualitative Results: Base + Novel Detections

- ViLD detects novel categories without sacrificing base performance.
- Robust across diverse scenes and object scales.
- Illustrates practical open-vocabulary utility.



Figure: LVIS/COCO/Objects365定性结果: ViLD检测新类且不降基类性能, 并可零微调迁移。

Direct Transfer: No Finetuning Needed

- PASCAL VOC 2007: AP50 $\approx$ 72.2; COCO val: AP $\approx$ 36.6; Objects365 v1: AP $\approx$ 11.8.
- Plug-and-play deployment by swapping text embeddings only.
- Robust even on lower-resolution images; occasional misses remain.



Figure: ViLD迁移至PASCAL VOC (分辨率低于LVIS训练集) 仍能检测; 第三图识别小瓶但漏检人物

Ablation: Distillation Loss and Weight Matter

- L1 distillation consistently improves rare-category APr over L2.
- Increasing λ boosts APr but can reduce AP on common/frequent classes.
- Deployment should tune λ to the target long-tail vs. head trade-off.

Table: Table 7: Hyperparameter sweep for visual distillation in ViLD.

Distill loss	Distill weight w	APr	APc	APf	AP
No distill	0.0	10.4	22.9	31.3	24.0
	0.5	13.7	21.7	31.2	24.0
L2 loss	1.0	12.4	22.7	31.4	24.3
	2.0	13.4	22.0	30.9	24.0
	0.05	12.9	22.4	31.7	24.4
	0.1	14.0	20.9	31.2	23.8

Interactive Detection and Prompt Engineering

- Multi-prompt ensembling slightly improves APr (15.7 $\rightarrow$ 16.1).
- Arbitrary texts enable attributes and fine-grained queries at test time.
- Supports systematic vocabulary expansion (e.g., colors, species) without retraining.

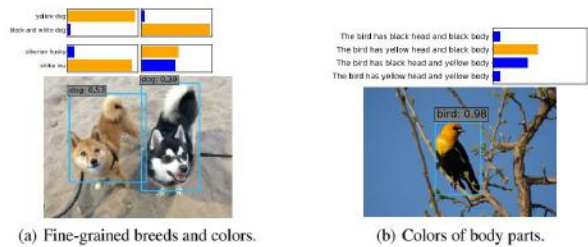


Figure: ViLD支持在线交互式目标检测: 任意文本细化细粒度类别与颜色属性

- Confusions among visually similar fine-grained categories.
- Aspect ratio distortion when resizing extreme crops degrades scores.
- Multiple objects inside a box can mislead the classifier.



Figure: CLIP在裁剪区域的典型错误: (a)预测与真值高相似; (b)缩放致纵横比失真; (c)受区域内他物干扰, 非整框目标。

- Classifier-on-crops may ignore box quality and rank poor boxes high.
- Geometric mean of class score and objectness improves rankings.
- Better aligns localization with recognition for higher-quality detections.



Figure: CLIP分数不反映框质量; Mask R-CNN框好但误判; 以CLIP分类得分与目标性分数几何均值重排, 优先高质类别正确框

- Masks can reflect low-level appearance over true object extent.
- Leads to fragmented or incomplete masks in zero-shot transfer.
- Suggests value in localization-aware scoring and improved mask priors.

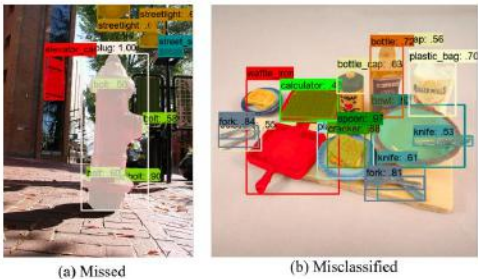


Figure: PASCAL VOC中掩码预测失败: 依赖低级外观而非语义

- Distills open-vocabulary vision-language knowledge into a fast detector.
- Aligns region features to text and image spaces; class-agnostic localization.
- Strong zero-shot/generalized detection; direct cross-dataset transfer via text swapping.
- Scales with stronger teachers; deployable without changing detector architecture.

- Better proposals and localization-aware scoring for unseen categories.
- Balancing text vs. image objectives; robust cropping pipelines.
- Richer prompt/attribute modeling; ethical deployment in sensitive domains.

- Thank you for your attention!
- Questions and feedback are welcome.

Acknowledgments

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